

**MARK SCHEME for the October/November 2009 question paper
for the guidance of teachers**

9702 PHYSICS

9702/22

Paper 22 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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- 1 (a) (i) *either* 1.55% *or* 1.6% ...(not 1.5 or 2) A1 [1]
(ii) *either* 1.09% *or* 1.1% ...(not 1.0 or 1) A1 [1]
- (b) answer of {(ii) + 2 × (i)} to any number of sig. fig.
either 4.2% *or* 4.3% A1 [1]
- (c) (i) *either* the value has more significant figures than the data
or uncertainty of ± 0.4 renders more than 2 s.f. meaningless) B1 [1]
(ii) uncertainty in $g = \pm 0.41 / \pm 0.42$ to any number of s.f. C1
 $g = (9.8 \pm 0.4) \text{ m s}^{-2}$ A1 [2]

[Total: 6]

- 2 (a) (i) e.g. (phase) change from liquid to gas / vapour
thermal energy required to maintain constant temperature B1 [1]
(*do not allow 'convert water to steam'*)
- (ii) e.g. evaporation takes place at surface B1
boiling takes place in body of the liquid B1
e.g. evaporation occurs at all temperatures B1
boiling occurs at one temperature B1 [4]
- (b) (i) volume = $(\frac{48}{4.5}) = 10.7 \text{ cm}^3$ A1 [1]
- (ii) 1 volume = $10.7 / (6.0 \times 10^{23})$
= $1.8 \times 10^{-23} \text{ cm}^3$ A1 [1]
2 separation = $\sqrt[3]{(1.8 \times 10^{-23})}$
= $2.6 \times 10^{-8} \text{ cm}$ A1 [1]

[Total: 8]

- 3 (a) (i) speed = 4.0 m s^{-1} ...(allow 1 s.f.) A1 [1]
(ii) $v^2 = 2gh$
= $2 \times 9.8 \times 1.96$ M1
 $v = 6.2 \text{ m s}^{-1}$ A0 [1]
(*use of $g = 10 \text{ m s}^{-2}$ loses the mark*)
- (b) correct basic shape with correct directions for vectors M1
speed = $(7.4 \pm 0.2) \text{ m s}^{-1}$ A1
at $(33 \pm 2)^\circ$ to the vertical A1 [3]
(*for credit to be awarded, speed and angle must be correct on the diagram – not calculated*)

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- 5 (a) (i) frequency f B1 [1]
(ii) amplitude A B1 [1]
(b) π rad or 180° (unit necessary) B1 [1]
(c) (i) speed = $f \times L$ B1 [1]
(ii) wave is reflected at end / at P B1
either incident and reflected waves interfere M1
or two waves travelling in opposite directions interfere M1
speed is the speed of incident or reflected wave / one of these waves A1 [3]

[Total: 7]

- 6 (a) total resistance in series = $2R$
total resistance in parallel = $\frac{1}{2}R$ M1
ratio is $2R / \frac{1}{2}R = 4$ (allow mark if clear numbers in the ratio) A0 [1]
(b) at 1.5 V, current is 0.10 A C1
resistance = $V/I = \frac{1.5}{0.1}$
= 15Ω A1 [2]
(use of tangent or any other current scores no marks)

(c)

	p.d. across each lamp / V	resistance of each lamp / Ω	combined resistance / Ω
series	1.5	15	30
parallel	3.0	20	10

- column 1 A1
columns 2 and 3: max 3 marks with -1 mark for each error or omission A3 [4]
(d) (i) ratio is 3(allow e.c.f.) A1 [1]
(ii) resistance increases as potential difference increases B1
increasing p.d. increases current B1
current increases non-linearly so resistance increases B1 [3]

[Total: 11]

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- 7 (a) *either* forms of same elementM1
or atoms / nuclei with same number of protonsA1 [2]
atoms / nuclei contain different numbers of neutrons
(use of 'element' rather than atoms / nuclei scores max 1 mark)
- (b) (i) decay is not affected by environmental factors B1 [1]
(allow two named factors)
- (ii) *either* time of decay (of a nucleus) cannot be predicted B1 [1]
or nucleus has constant probability in a given time
- (c) $^{185}_{75}\text{Re}$ B1
either $^0_{-1}\text{e}$ *or* $^0_{-1}\beta$ B1 [2]

[Total: 6]